

Peter Lee

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EDUCATION

University of Toronto, St. George

Toronto, ON

Honours Bachelor of Science

Sept. 2024 – April 2028

- **Computer Science Specialist** (with **Co-op**)
- **GPA:** 4.0, **Scholarships:** Woodsworth College Admissions (\$10000), Dorothy Helen Mcrobb (\$2250)
- **Coursework:** Software Design, Data Structures & Algorithms, Software Tools & Systems Programming, Databases, Computer Organization

EXPERIENCE

Trustworthy Machine Intelligence

Oct. 2025 – Present

Software Engineer

Toronto, Ontario

- Architected a 3-tier cloud pipeline (Next.js, FastAPI, GPU inference server) for a context-aware lecture captioning system, ingesting PDF course materials to guide real-time transcription with end-to-end latency under 5 seconds.
- Engineered a WebSocket streaming layer using 3 second audio chunking to continuously proxy live audio to an in-house GPU inference server, enabling low-latency captions and post-session transcript generation at scale.
- Designed a PostgreSQL schema and async storage system handling concurrent PDF and audio uploads, securing user sessions with JWT authentication using Supabase.

Code Ninjas

Feb. 2022 – Aug. 2024

Coding Instructor

Mississauga, Ontario

- Instructed **150+** students aged 8–18 in game development using Unity (C#), JavaScript, and Lua across 2 years.
- Authored **16** pages of structured Roblox curriculum documentation, standardizing lesson delivery for students.

PROJECTS

GitChores | *Java, Swing, AWS*

- Built a Java desktop app with a GitHub-style contribution heatmap for roommate chore tracking, applying Clean Architecture and SOLID principles across a fully layered codebase.
- Integrated AWS DynamoDB for NoSQL persistence and AWS Cognito for OAuth authentication, supporting multi-user session management.
- Automated CI/CD via GitHub Actions, executing **210** pipeline runs and resolving **26** build failures to maintain a stable main branch.

Business Flight Visualizer | *Python, pycountry, networkx, pandas, dash*

- Built a graph-based global meeting optimizer in Python using DFS over a network of **10,000+** airport nodes to compute minimum-travel central meeting locations.
- Achieved **0.09ms** query time on average for a full graph traversal, benchmarked across **20,000** runs using timeit.

CASCH Research | *Python, pandas, scikit-learn*

- Analyzed **10,000+** survey observations to model predictors of employee happiness using random forests.
- Achieved **84%** test-set accuracy by evaluating the model performance using confusion matrices and key classification metrics.

PROtectABot4000 | *discord.py*

- Awarded **\$5,000** in funding from the DMZ Startup Incubator (Top 4) for building a Discord bot that utilizes keyword flagging to detect and analyze political bias in real-time server message streams.

TECHNICAL SKILLS

Languages: Python, C, Java, TypeScript, JavaScript, C#, HTML5, CSS

Libraries/Frameworks: React, Next.js, FastAPI, Node.js, Express.js, React Native, Tailwind CSS, pandas, scikit-learn

Databases & Cloud: PostgreSQL, Supabase, MongoDB, Amazon Web Services (AWS)

Tools & Testing: Claude Code, Docker, Linux, Git, GitHub Actions (CI/CD), WebSockets, REST APIs, Jest, pytest, Playwright